

Definition. Let f be a function defined at a . The n^{th} order Taylor polynomial for f with its center at a , denoted p_n is

$$p_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k.$$

1. Complete the following.

a. Find a Taylor polynomial of order $n = 3$ for $\ln(x)$.

$$f'(x) = \frac{1}{x}, f''(x) = -\frac{1}{x^2}, f'''(x) = \frac{2}{x^3}.$$

$$p_3(x) = \ln(a) + \frac{1}{a}(x-a) - \frac{1}{2a^2}(x-a)^2 + \frac{1}{3a}(x-a)^3$$

b. Use the Taylor polynomial from above to approximate $\ln(1.07)$.

Let $a = 1$ (the center is 1 since $1 \approx 1.07$). So then $f(1) = 0, f'(1) = 1, f''(1) = -1, f'''(1) = 2$, and we have

$$\begin{aligned} p_3(x) &= (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 \\ \Rightarrow p_3(1.07) &= .067664\bar{3} \end{aligned}$$

c. Find the absolute error of the approximation above.

$$\ln(1.07) \approx .067659, \text{ so then error} = |\ln(1.07) - p_3(1.07)| \approx 0.000005$$

2. Use a Taylor polynomial of order $n = 5$ to approximate $\cos(0.1)$.

Let $f(x) = \cos(x)$, and $a = 0$ since 0.1 is close to 0 and $\cos(0) = 1$. $f'(x) = -\sin(x)$, $f''(x) = -\cos(x)$, $f^{(3)}(x) = \sin(x)$, $f^{(4)}(x) = \cos(x)$, and $f^{(5)}(x) = -\sin(x)$. So then $p_5(x) = 1 + 0 - \frac{x^2}{2} + 0 - \frac{x^4}{24} + 0 = 1 - \frac{x^2}{2} - \frac{x^4}{24}$. So then $p_5(.1) \approx .9949$.

3. Approximate $\frac{1}{\sqrt{4.08}}$ using a Taylor polynomial of order $n = 2$.

We'll use $f(x) = \frac{1}{\sqrt{4+x}}$ with $a = 0$ since $\frac{1}{\sqrt{4}} = \frac{1}{2}$. $f'(x) = -\frac{1}{2}(4+x)^{-3/2}$ and $f''(x) = \frac{3}{4}(4+x)^{-5/2}$. So then $p_2(x) = \frac{1}{2} - \frac{1}{2}(4+x)^{-3/2}x + \frac{3}{8}(4+x)^{-5/2}x^2$, and $p_2(.08) \approx .511719$ ($\frac{1}{\sqrt{4.08}} \approx .49507$).